

SIMRAN SAINAND REVANKAR

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Research Experience

Project Associate – I | Indian Institute of Technology Palakkad, India

Department of Electrical Engineering

Mar 2025 – Jun 2026

Project: *Stochastic Resonance Based Detectors Using Thin-Film Integrated Ferroelectrics*

Funding: Core Research Grant, Science and Engineering Research Board (SERB) | **Approved funding: INR 73.73 Lakh**

Project duration: Dec 2022 – Dec 2025

- Developed reproducible **PVDF-TrFE thin-film fabrication** workflows using spin coating, Langmuir–Blodgett deposition and drop casting.
- Investigated the influence of **solution chemistry, solvent composition, drying and annealing conditions** on thin-film properties and electrical behaviour.
- Fabricated **ferroelectric capacitor devices** and investigated processing–structure–property relationships through electrical measurements.
- Performed **P–E hysteresis and PUND measurements** using a Radiant Technologies ferroelectric tester.
- Gained exposure to thin-film deposition workflows including sputtering and thermal evaporation within device-fabrication processes.
- Analysed experimental datasets and investigated relationships between processing conditions, thin-film properties and device performance.

Independent Researcher | JSS Science & Technology University, India

Department of Chemistry

Nov 2024 – Mar 2025

Research focus: Perovskite materials, interfaces and functional polymer systems

- Investigated Cu-based and Rb-based perovskite systems for energy-harvesting applications.
- Examined how **composition and nanoscale interfaces** influence material and optoelectronic behaviour.

Graduate Researcher | JSS Science & Technology University, India

Department of Chemistry

Dec 2023 – Sep 2024

Research focus: Halide perovskites, polymer nanocomposites and functional devices

- Synthesized and characterized **halide perovskite materials**, including lead-based and lead-free compositions, and developed perovskite/polymer nanocomposites for functional devices.
- Investigated relationships between **material composition, morphology, interfaces and device performance**.
- Fabricated flexible triboelectric devices and evaluated their electrical response under different material compositions and processing conditions.
- Characterized materials using UV–Vis, FTIR, photoluminescence, contact-angle measurements, XRD and SEM, with AFM data interpretation.
- Published research on fluorescent perovskite/polymer nanocomposites for self-powered UV sensing.

Research Intern | CSIR–Central Food Technological Research Institute, India

Department of Plantation Product, Spices and Flavour Technology

Nov 2023 – Dec 2023

- Optimized curcumin extraction using spectroscopic and chromatographic analysis.
- Performed characterization using 1D NMR, FTIR, UV–Vis and HPLC.

Research Capabilities

Perovskite materials	Halide perovskites; lead-based and lead-free compositions; perovskite/polymer nanocomposites; interfaces; structure–property relationships
Thin-film processing	Solution preparation; spin coating; Langmuir–Blodgett deposition; drop casting; solvent processing; thermal annealing; substrate preparation
Device fabrication	Functional thin-film devices; ferroelectric capacitors; triboelectric nanogenerators; electrode/device assembly
Materials characterization	XRD; FTIR; Raman; UV–Vis; photoluminescence; contact angle; SEM; AFM data interpretation
Electrical characterization	P–E hysteresis; PUND; electrical response analysis; device performance evaluation
Data analysis	Python; OriginPro; ImageJ; Gwyddion; quantitative data analysis and visualization

Publications

- S. S. Revankar, Y. V. Lokesh, A. S. Manjunatha, . . . , B. S. Madhukar, “Fluorescent CsPbBr₃@Cs₄PbBr₆/PU polymer nanocomposite-based triboelectric nanogenerator for self-powered UV sensing,” *Materials Today Chemistry*, 44 (2025), 102601. **First author**. DOI: 10.1016/j.mtchem.2025.102601
- N. Manjushree, Y. V. Lokesh, K. R., . . . , **S. S. Revankar**, . . . , B. S. Madhukar, “Characterization and DFT investigation of CuWO₄@CMC/PVA composite with down conversion ability,” *Journal of Molecular Structure*, 1325 (2025), 140780. **7th author**. DOI: 10.1016/j.molstruc.2024.140780
- K. M. Gowda, Y. V. Lokesh, U. G. P. . . . , **S. S. Revankar**, . . . , B. S. Madhukar, “Flexible, lead-free Cs₃Bi₂Br₉@EVA nanocomposite triboelectric nanogenerator for energy harvesting and tactile sensing,” *Emergent Materials*, 8 (2025), 5827–5842. **7th author**. DOI: 10.1007/s42247-025-01170-0

Cumulative Journal Impact Factor: 14.6

Education

M.Sc. Chemistry | JSS Science & Technology University, India 2022 – 2024
CGPA: 9.50/10 | GPA: 3.95/4.0

Thesis: “Flexible and fluorescent CsPbBr₃@Cs₄PbBr₆/PU polymer nanocomposite as triboelectric nanogenerator for self-powered UV sensor.”

B.Sc. Chemistry, Zoology & Biotechnology | Maharani Science College for Women, India 2019 – 2022
CGPA: 8.90/10 | 3.7/4.0

Advanced Research Training

- **Nanostructured Optoelectronic Materials** – NIT Warangal, India, 2025.
- **Nanoelectronics & Circuits** – IIT Palakkad, India, audited, 2025.

Research Funding & Recognition

- **KSCST Student Research Project:** “Lead-free Perovskite/Polymer Nanocomposite as Triboelectric Nanogenerator.” Funded by the **Karnataka State Council for Science and Technology (KSCST)**, **Government of Karnataka**; Ref. 47S_MSC_0097; Approved funding: INR 7,000.
- **Best Oral Presentation Award**, International Conference on Polymers for Sustainable Development, 2025.

Selected Presentations

- Poster Presentation – 17th International Postgraduate Summer School on Green Chemistry, Venice, Italy, 2025.
- Oral Presentation – International Conference on Nanostructured Materials and Nanocomposites, 2024.
- Poster Presentation – International Symposium on Nanomaterials for Health and Energy, 2024.
- Poster Presentation – 47th Student Project Program, State-Level Competition, 2024.

References

Dr. Arvind Ajoy

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Dr. Madhukar B. S.

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Dr. Sachin R. Chaudhari

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